ECG Heart Rate Monitoring Sensor Using MATLAB

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Abstract: **This paper presents the design and implementation of an Arduino-based ECG heart rate monitoring system. The proposed system uses an ECG sensor module to acquire cardiac signals, which are then processed and analyzed in real time using an Arduino microcontroller. The signal analysis enables accurate determination of heart rate and effective monitoring of cardiac activity. The extracted heart rate data is displayed on an LCD screen and can also be transmitted wirelessly for remote health monitoring. The developed system provides a cost-effective, compact, and reliable solution for continuous cardiac observation, making it suitable for wearable healthcare and telemedicine applications. Experimental evaluation demonstrates that the system achieves stable performance and high accuracy in heart rate detection when compared to standard clinical monitoring devices.**

Keywords — ECG, Heart Rate Monitoring, Arduino, AD8232, Biomedical Signal Processing, Analog Signal Acquisition, Serial Communication, Real-time Monitoring, Wearable Health Device, Noise Filtering, Signal Conditioning, Data Visualization, IoT-based Health Monitoring, Patient Monitoring System

I. INDRODUCTION

The ongoing monitoring of heart activity has increasingly become significant in early cardiovascular disease detection and prevention, which is still among the major causes of death globally. The conventional hospital-based ECG[1][16] device is, although accurate, bulky, costly, and not user-friendly for ongoing personal health monitoring. With the quick expansion of low-cost microcontrollers and biomedical sensors, there exists a greater incentive to create a low-cost, portable, and real-time ECG heart rate monitoring system. Through the utilization of Arduino[14] along with an ECG sensor[1] like the AD8232[2] , it is feasible to read and analyze cardiac signals in an easier way, thus opening up healthcare to greater accessibility, particularly in remote and resource-constrained settings. The research is driven by necessity to offer low-cost, wearable, and easy.

A number of researchers have developed ECG-based heart[1] monitoring systems based on various sensor modules and microcontrollers. Previous studies show the viability of applying the AD8232[2] sensor with Arduino[14] to capture cardiac signals and compute heart rate in real time. Other research works have combined ECG acquisition[13][16] systems with MATLAB, Python, or IoT [8] platforms for enhanced analysis, wireless data, and cloud-based monitoring. Whereas these methods have yielded encouraging results, the majority of them are hampered by high installation cost, non-portability, or computational intensity in signal processing methods.

The primary objective of this research work is the design of a low-cost, portable ECG[12][16] heart rate monitoring system based on Arduino[14] and AD8232[2] sensor. The system outlined in this work is anticipated to efficiently capture ECG signals[16], process them in real time, and provide accurate heart rate estimation with noise and motion artifact rejection. For this, the Arduino microcontroller is applied for data acquisition and signal condition, and serial communication[11] for the graphical interface or MATLAB visualization.

II. PROBLEM STATEMENT

Cardiovascular disorders are a major cause of mortality globally, and continuous monitoring of heart rate is crucial for early detection and prevention of cardiac complications. Traditional ECG devices are large, costly, and restricted to clinical settings, which hinders their usefulness for real-time monitoring. Patients with no convenient access to neighboring or rural healthcare facilities experience delay in diagnosis of heart abnormalities. Hence, there is an urgency for a miniaturized, low-cost, and portable ECG heart rate monitoring sensor that can sufficiently obtain, process, and transmit ECG signals for real-time monitoring.

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III. OBJECTIVE

The primary objective of this research is to design and develop an effective ECG heart rate monitoring sensor system capable of accurately measuring and recording cardiac activity. The system is intended to provide continuous monitoring of heart activity at a low cost using Arduino-based processing in a portable setup. In addition, the objective is to ensure accurate heart rate measurement, support early detection of cardiac abnormalities, and enable applications in telemedicine, remote healthcare monitoring, and personal healthcare management..

ECG SIMULATOR

Basically, the ECG simulator is intended to mimic normal ECG waveforms for various leads, as well as mimic abnormal ECG rhythms in the form of various arrhythmias. The ECG simulator, developed with the use of MATLAB, is able to mimic normal ECG waveforms for Standard Lead II leads. Some of the advantages that come with the use of an ECG simulator include saving time that could have been used in data acquisition, as well as the flexibility it brings in terms of not requiring an ECG recording tool in order to perform an analysis on normal and abnormal ECG signals. With the ECG simulator, therefore, any ECG signal waveform can easily be investigated.

Key Characteristics of ECG Waveforms:

Figure 1 above illustrates the general scalar leads of an electrocardiograph with the prominent features of the waveform, which includes the P, Q, R, S, and T waves. An ECG is a periodic signal; the fundamental frequency depends on the heart rate. The ECG-signal meets Dirichlet's requirements; thus, the Fourier series can be applicable. On detailed examination, the cycle of an ECG could be composed of other functions like triangles and sinusoids. Every particular ECG feature can be defined by the shifted functions of these basic components. These particular components of an ECG could be defined by functions that are either triangular or sinusoidal. The QRS complex, Q segment, or S segment can be defined by the triangular functions, while the P, T, or U waves are defined by sinusoidals.

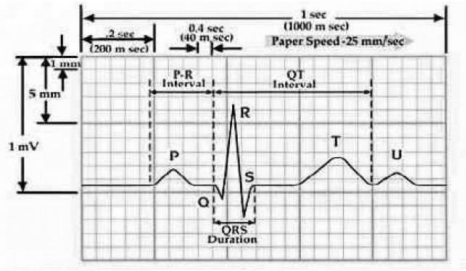


Fig.1 one-type ECG signal tracking

ECG SIGNAL PROCESSING

ECG signal processing is a multi-step process that converts raw cardiac signals into clinically relevant information for diagnosis and monitoring. It starts with preprocessing, in which baseline wander, power-line interference, and motion artifacts are eliminated by digital filters, wavelet denoising, or adaptive techniques. Band-pass and notch filtering subsequently maintain the diagnostically important frequency components of the signal and reject noise.Following purification, feature extraction algorithms like QRS detection, calculation of RR intervals, and analysis of heart rate variability (HRV) are used for finding morphological and temporal properties of the P-QRS-T waveform.

These features are used as inputs for decision-making modules that may be classical rule-based systems, statistical classifiers, or even sophisticated machine learning and deep learning-based models for detection the of arrhythmia and prediction of disease. Applications include the use in hospital-based diagnosis systems and wearable and telemedicine platforms to facilitate continuous monitoring and early detection of cardiovascular abnormalities. Current research is aimed at resilient processing under noisy conditions, multimodal biosignal integration, and the development of explainable AI solutions to improve reliability and clinical acceptance.

Table 1. Parameter Table

Parameter	Typical Value
Electrode Type	Ag/AgCl
Input impedance	> 10 MΩ
CMRR	> 80–100 dB
Analog bandwidth	0.05 – 150 Hz (typical)
Sampling frequency	250 – 1000 Hz
ADC resolution	12 – 16 bits
Total gain	500 – 2000×
Baseline cutoff (HP)	0.05 – 0.5 Hz
Notch	50/60 Hz

ECG READINGS

Normal Heart Rate (Normocardia):

A normal adults heart rate is ranges between 60 and 100 beats per minute (bpm), and it is recorded on the electrocardiogram (ECG) as normally spaced P-QRS-T complexes with consistent shape. The RR intervals are also equal, showing constant conduction via the sinoatrial (SA) node and cardiac

conduction system. Normocardia is seen as an indicator of well-balanced autonomic control and normal cardiac rhythm, which serves as a reference point with abnormal states.

Slow Heart Rate (Bradycardia):

Bradycardia is a rate less than 60 bpm and is characterized on the ECG by extended RR intervals. Although sinus bradycardia can be physiologically normal during sleep or in athletes, it can also be caused by conduction system disease, hypothyroidism, or medication side effects like beta-blockers. In pathological bradycardia, one may present with dizziness, fatigue, or syncope, and sinus pauses or atrioventricular block may be evident on the ECG.

Fast Heart Rate(Tachycardia):

Tachycardia is defined as a heart rate of more than 100 bpm, with decreased RR intervals on the ECG. Sinus tachycardia usually maintains a normal P-QRS-T morphology but at an increased rate, usually caused by stress, fever, or exertion. On the contrary, supraventricular and ventricular tachyarrhythmias can change waveform morphology, causing abnormal positioning of the P-wave, broadened QRS complexes, or unsynchronized rhythms. Chronic tachycardia can decrease cardiac efficiency and predispose to arrhythmias, myocardial ischemia, or sudden cardiac death.

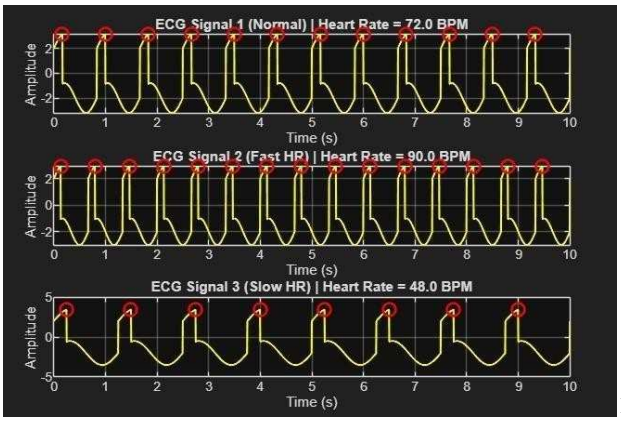


Fig 2 ECG Heart rates

COMPARISION OF DIFFERENT PROCESSING TECHNIQUES

FREQUENCY-DOMAIN METHODS:
Frequency-domain methods applied include Fourier and power spectral analysis, which are used for HRV measurement. They are good at giving information about autonomic regulation but are poor for direct detection of the R-peak, as they do not have high temporal resolution.

TIME-DOMAIN METHODS:
Traditional time-domain methods have been based on thresholding and differentiation-based algorithms for R-peak detection, e.g., the Pan–Tompkins algorithm. These methods are computationally efficient and appropriate for real-time embedded platforms, but their performance is sensitive to noisy conditions or motion artifacts.

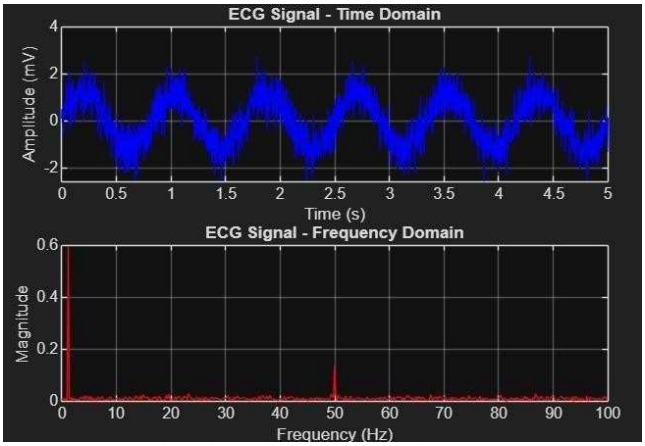


Fig 3 Time and frequency domains

IV. RESULTS

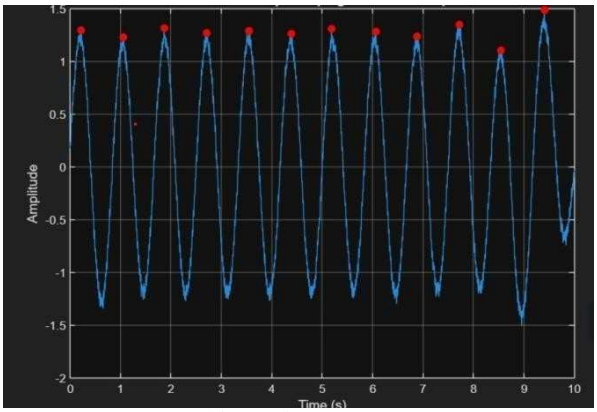


Fig. 3. ECG with R-peaks(Avg HR = 71.9 BPM)

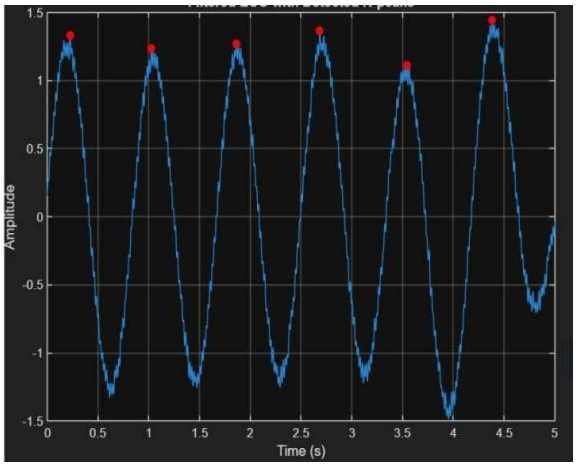


fig 4. Filtered ECG with detected R-peaks

V. Conclusion

Development of an ECG heart rate monitoring system based on the AD8232 sensor and Arduino Nano reflects a cost-efficient and reliable method of acquiring and processing cardiac signals. The system successfully records real-time ECG signals, processes them via the Arduino, and offers reliable feedback on heart activity. Due to its miniaturized size, low energy consumption, and portability, the setup is well-adapted to regular health monitoring both in non-clinical and clinical settings.

The lead-off detection inclusion makes the system reliable through the alert of improper electrode attachment, thus enhancing the usability and accuracy of the system. Furthermore, real-time visualization and analysis are by being facilitated through design, which makes it applicable in the early stages of a project, the swarm analysis of abnormalities in heart function. The findings indicate the potency of the system as an effective and useful tool for patients, researchers, and professionals..

In conclusion, this study highlights the importance of Integration of Biomedical Sensors and Microcontrollers for Practical Healthcare Use Cases. The ECG Monitoring System ensures it provides a platform for future developments, including wireless data transfer functionality; cloud connectivity capabilities; smart signal functionality analysis. These developments would further cement the stance on portable ECG machines in telemedicine, remote health Monitoring, and preventive health care.

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